The Knee 52 (2025) 99–107

Contents lists available at ScienceDirect

The Knee

journal homepage: www.elsevier.com/locate/thekne

Association between intra and inter-limb strength asymmetry with sprint kinematics and force-velocity profile in youth team athletes

İzzet İnce [a,*], Erkan Tortu [b]

[a] Faculty of Sports Sciences, Department of Coaching Education, Ankara Yıldırım Beyazit University, Ankara Türkiye
[b] Faculty of Sport Sciences, Department of Coaching Education, Trabzon University, Trabzon, Türkiye

ARTICLE INFO

Article history:
Received 11 April 2024
Revised 25 September 2024
Accepted 23 October 2024

Keywords:
Maturation
Lower limb asymmetry
Sprint kinetics
Force–velocity profile
Soccer

ABSTRACT

Background: Maximal force plays a pivotal role in enhancing performance across various dynamic sports, particularly in sprinting biomechanics. However, muscle strength asymmetry among major muscle groups like the hamstrings and quadriceps may hinder sprint performance and raise injury risks. This study examines how intra- and inter-limb strength asymmetry relates to sprint kinematics and the power-force–velocity profile in youth athletes.

Methods: Seventy-four youth athletes from football, volleyball, and handball completed two testing sessions, 48 h apart. Assessments included anthropometric measurements, sprint tests, and isokinetic dynamometer strength evaluations at $60°.s^{-1}$ and $180°.s^{-1}$. Sprint kinematics were analyzed using the Optojump Next system, and force–velocity profile variables were determined via Samozino's method. Asymmetry analysis used a standard percentage difference equation.

Results: Weak to moderate correlations ($r = -0.46$ to 0.45) were found between strength asymmetry and sprint parameters, suggesting limited impact of asymmetry on sprint kinematics and kinetics. Although most participants showed asymmetry levels below the 15% threshold deemed functionally significant, outliers had asymmetry values up to 42%, indicating considerable variability.

Conclusions: Lower extremity strength asymmetry moderately affects sprint kinematics and force–velocity profiles in young team athletes, challenging traditional views. Further research is needed to explore the mechanisms between strength asymmetry and sprint performance, aiming to inform targeted training interventions that optimize performance and reduce injury risks in youth athletes.

© 2024 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

1. Introduction

Attaining an acceptable performance threshold relies inherently on the development of increased force and motor proficiency, particularly emphasizing key muscle groups such as the hamstrings and quadriceps [1,2]. In current literature, it is proposed that disparities in muscle strength distribution among major muscle groups, such as the hamstrings and

* Corresponding author at: Ankara Yıldırım Beyazıt University, Ayvalı mh. Gazze Cd. No:7, Keçiören/Ankara Türkiye.
E-mail addresses: izzetince@aybu.edu.tr (İ. İnce), erkantortu@trabzon.edu.tr (E. Tortu).

https://doi.org/10.1016/j.knee.2024.10.020
0968-0160/© 2024 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

quadriceps, may detrimentally influence sprint performance [3–5]. Consequently, the identification of asymmetries in lower limb muscles emerges as a crucial factor potentially impacting sprinting biomechanics, thus warranting further investigation.

Recent research in the field of applied biomechanics has highlighted certain mechanical descriptors of running gait that exhibit significant influence on sprint performance, particularly at maximal and submaximal speeds [6]. Notably, studies have underscored the importance of parameters such as stride length and frequency, relative support time, and the time ratio between the braking and propulsion phases during the support [2,7]. These parameters are intricately linked with the structure of the stride and have been shown to play a crucial role in optimizing sprinting biomechanics and enhancing overall performance [2]. The general mechanical capacity to generate horizontal external force during sprinting is illustrated by the linear force–velocity (FV) relationship [8,9]. In conjunction with the insights from isokinetic knee extension and flexion strength studies, which have demonstrated a strong association with improved sprint performance [10–12], the mechanical determinants identified through FV profiling, as proposed by Samozino et al. [13], have garnered significant attention. This methodology offers a means to quantify the mechanical aspects underlying sprint performance and has become a focal point in recent investigations aimed at understanding the intricate interplay between power, velocity, and force in sprinting [8,9].

Significantly, as investigations explore further into the biomechanical intricacies of sprinting, there has been a notable increase in interest concerning the association between inter and intra-limb asymmetries in muscle strength and power, and their implications for athletic performance [14]. One study concluded that there is no direct relationship between athletes' overall asymmetry in limb strength and sprint performance, suggesting that strength imbalances are joint- and task-specific rather than dependent on limb dominance [6]. Another study found significant positive correlations between net ankle work in sprinting and peak force and peak power during strength diagnostic jump tests, indicating an interaction between asymmetry in sprint performance and lower limb strength [15]. Conversely, a study on male volleyball players found no significant relationship between sprint and isokinetic knee strength at different angular velocities [1], while research on female sprinters detected differences in kinetic parameters and a negative correlation between asymmetry and certain kinetic parameters, though no direct correlation with sprint performance was apparent [16]. Thus, the literature elucidates a perplexing paradox concerning the influence of asymmetries on performance across diverse lower body strength assessments, compelling a deeper exploration of their implications for athletic excellence. Optimizing sprint performance depends heavily on biomechanical efficiency, particularly in key muscle groups such as the hamstrings and quadriceps [1,2]. Research increasingly suggests that asymmetries in lower limb muscle strength may disrupt the mechanics crucial for sprinting, yet findings remain inconsistent [3–5]. While some studies indicate that strength imbalances negatively impact performance metrics such as stride length and velocity, others suggest that these effects may be joint- and task-specific [6,14]. The relationship between muscle asymmetry and sprint kinematics, particularly in youth athletes, remains inadequately explored.

Biomechanical parameters such as stride length, step velocity, and the force–velocity (FV) profile are well-established determinants of sprint performance [6,7]. The ability to generate horizontal force, as captured by FV profiling, is critical in achieving maximum sprint speeds [8,9]. Despite the importance of these parameters, it is unclear whether strength asymmetries in key muscle groups impair an athlete's capacity to generate optimal force and velocity during sprinting. Addressing this gap is essential, as early intervention in youth athletes could yield significant performance benefits.

This study builds on previous work by examining the influence of intra- and inter-limb asymmetries on both kinematic and kinetic aspects of sprinting. We anticipate that strength asymmetries will manifest in altered biomechanical profiles, potentially limiting force production and velocity. Furthermore, we explore the potential for these asymmetries to impact sprinting efficiency, with the aim of providing new insights into the interplay between muscle strength and sprint performance. By doing so, this research aims to offer valuable contributions to the development of more effective training protocols for youth athletes.

2. Methods

2.1. Experimental design

In this cross-sectional study, G-Power 3.1.9.7 analysis, with parameters tails = two, pH1 (r) = 0.343, $\alpha = 0.05$, power (1-β) = 0.80, and pH0 = 0, indicated a need for 64 participants. To account for potential dropout or health-related issues, approximately 20% more participants than the targeted number were recruited [17], resulting in an initial enrolment of 77 athletes. However, three athletes discontinued participation due to illness or pain, leading to a final sample size of 74 athletes. Effect size (r) values analyses were determined based on a previous study. [18]. Participants underwent testing twice with a 48-hour interval between sessions during the data collection phase. Athletes were categorized into three distinct groups for the testing procedures. Each group completed two test sessions on separate days. The two test sessions were structured as follows: (1) Evaluation of body composition, anthropometric measurements, and a 30-meter sprint test; (2) Isokinetic dynamometer strength assessment (Figure 1). Since the participants, aiming to contribute to the talent pool of budget sports clubs, had previously undergone these tests at regular intervals no familiarization sessions were conducted. To minimize

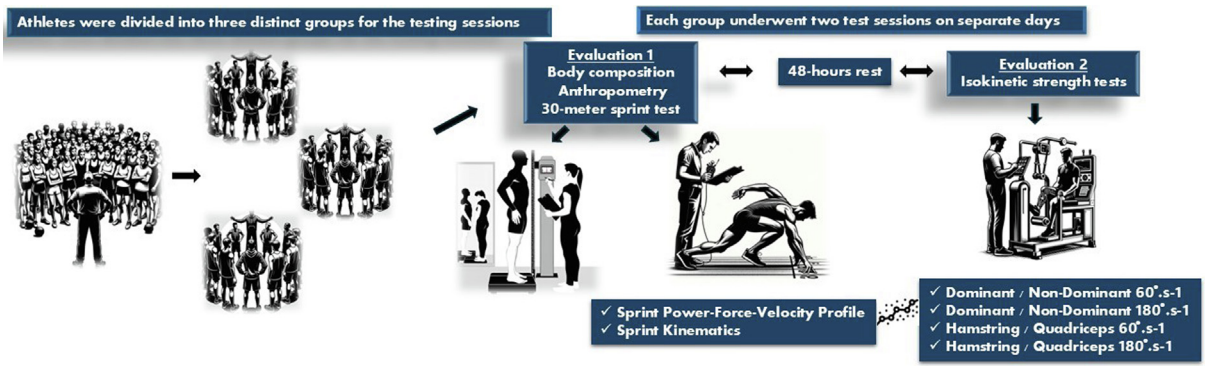


Figure 1. Study design: schematic representation of study design.

potential test–retest and device errors, the primary investigator conducted the sprint test using consistent test devices. The isokinetic strength assessment was conducted by the secondary investigator, with the assistance of a physiotherapist.

2.2. Participants

This study included 74 young athletes: 34 football, 23 volleyball, and 17 handball players. Their anthropometrics, body compositions, and maturity offset values are summarized in Table 1. Football players trained five days a week, volleyball players a minimum of four sessions, and handball players engaged in a competitive team with similar training frequency. Exclusion criteria comprised no musculoskeletal or neurological injuries, no history of knee injuries, and absence of conditions aggravated by the protocol. Participants and their guardians were fully informed about the study's objectives, methods, potential benefits, and risks. Informed consent was obtained, and ethical approval was granted by the Central University Health Sciences Ethics Committee in compliance with the 2008 Helsinki Declaration.

2.3. Anthropometric characteristics and maturity offset

Body mass was measured with a Tanita body fat analyzer (Tanita HD–318, Tanita Corporation, Japan), and height with a 2-meter-long wall-mounted stadiometer (Mesilife, model SW-G06B). These parameters, along with birth and measurement dates, were recorded for maturity offset calculation, estimating proximity to peak height velocity (PHV) as a proxy for biological maturation in male participants using the Modified Maturity Offset Prediction Equations [19] which is expressed as follows.

$$Maturity\ \ offset\ (years) = -7.999994 + (0.0036124)x(age\ \ x\ \ body\ \ height) \quad (1)$$

This equation provides an approximation of the time (in years) until, or since, an individual's predicted PHV, thereby elucidating their biological maturity status.

2.4. Isokinetic measurements

Isokinetic knee extensor strength was assessed using a Cybex HUMAC Norm (CSMi, Stoughton, MA, USA) at two speeds: 60°.s-1 for five repetitions and 180°.s-1 for 15 repetitions. Participant details were recorded, and the dynamometer was calibrated according to manufacturer guidelines. A standardized warm-up on a cycle ergometer preceded testing. Participants

Table 1
Anthropometric, body composition and maturity offset parameters.

Variables	Football Players (n = 34)		Volleyball Players (n = 23)		Handball Players (n = 17)	
	X	**SD**	**X**	**SD**	**X**	**SD**
Age (years)	16.14	0.54	16.50	1.81	16.78	1.53
Body Mass (kg)	69.51	6.34	73.79	8.93	70.18	4.58
Body Height (cm)	175.36	3.83	187.83	7.74	171.74	3.74
Body Fat (%)	11.42	3.08	14.08	2.13	13.29	2.87
Training Years (years)	5.92	1.22	5.71	1.19	3.59	0.74
Maturity Offset (years)	1.74	0.97	1.86	0.92	1.79	1.03

X: Mean; SD: Standart deviation; n = Sample size.

were seated with hips flexed at 90°, knees aligned with the dynamometer axis. Stabilization straps secured participants during testing. They performed three maximal contractions for each velocity and contraction type. Absolute and relative peak moments were determined for dominant and non-dominant legs. The highest valid value was chosen from two trials per velocity, with a three-minute rest between trials and a five-minute rest between velocity changes [20,21]. ICC values were 0.91 (95% CI: 0.88–0.94) at 60°.s-1 and 0.92 (95% CI: 0.90–0.94) at 180°.s-1, indicating high reliability.

2.5. Sprint kinematics

Participants completed a standardized warm-up: 3 min of light jogging followed by 10-second static stretches targeting hamstrings, quadriceps, and calf muscles, repeated 3 times. They then performed sprint-specific drills including "A march," "A skip," power skips over 10 m, and wall drive exercises to enhance sprint performance. Kinematic variables (including 5-meter sprint time (S5m), 10-meter sprint time (S10m), 20-meter sprint time (S20m), 30-meter sprint time (S30m), Left contact time (LCT), Right contact time (RCT), Left flight time (LFT), Right flight time (RFT), Left step length (LSL), Right step length (RSL), Left step velocity (LSV), and Right step velocity (RSV)) were recorded for the initial 30 m of each sprint using the Optojump Next system (Microgate, Bolzano, Italy). This optical system calculates step length, frequency, contact time, and flight time. Two trials were conducted for each participant to assess reliability, yielding an ICC of 0.89 (95% CI: 0.86–0.92), indicating consistent measurements.

2.6. Sprint force-velocity profile

Participants underwent 5, 10, 20, and 30-meter sprint tests on a parquet surface using Microgate's photoelectric gates, following established protocols [22]. Each player had two attempts with a two-minute rest interval between to prevent performance decline. Air pressure and temperature were measured using a barometer and temperature monitor (TFA 3006.42) before the windless tests. Horizontal FV profile variables were computed following guidelines by [13], using inputs including body mass, height, ambient temperature, ambient pressure, and results from the sprints. Prior to calculation, the single exponential equation [13] was used to model the horizontal velocity–time curve during maximal acceleration.

$$v_H(t) = VH_{max} \times \left(1 - e^{\frac{-t}{TAU}}\right) \tag{2}$$

In the above formula, VH_{max} represents the maximal velocity reached at the end of the acceleration (expressed as "ms^{-1}") and TAU represents the acceleration time constant (expressed as "s"). Following the derivation of the formula, horizontal acceleration (a_H) was computed as:

$$a_H(t) = \frac{VH_{max}}{TAU} \times e^{\frac{-t}{TAU}} \tag{3}$$

Upon calculating a_H using the formula above, the distance covered by the athlete over time (horizontal position, $x(t)$) was determined as:

$$x(t) = VH_{max} \times \left(t + TAU \times e^{\frac{-t}{TAU}}\right) - VH_{max}TAU \tag{4}$$

After obtaining $x(t)$ as a function of time, the net horizontal antero-posterior ground reaction force (F_{HGRF}) applied to the athlete's center of mass (CM) over time was computed as:

$$F_{HGRF}(t) = m \times a_H(t) + F_{aero}(t) \tag{5}$$

In the above formula, m represents the body mass (in "kg"). The aerodynamic drag force (F_{aero}) encountered by the athlete, dependent on wind velocity (V_w) during the sprint, was calculated as:

$$F_{aero}(t) = CF_{aero} \times (v_H(t) - V_w)^2 \tag{6}$$

The athlete's aerodynamic friction coefficient (FC_{aero}) required for the calculation above was determined as:

$$CF_{aero} = 0.5 \times \rho \times A_f \times CD \tag{7}$$

The air density (ρ) necessary for calculating CF_{aero} was determined as:

$$\rho(kg/m^3) = p_0 \times \frac{P_b}{760} \times \frac{273}{273 + C^{\circ}} \tag{8}$$

The athlete's frontal area (A_f) required for calculation was determined as:

$$A_f = \left(0,2025 \times h^{0,725} \times m^{0,425}\right) \times 0,266 \tag{9}$$

In the above formulas, h is the participant's stature (in "m") while P_b is the barometric pressure (in "Torr"). Additionally, CD represents the drag coefficient (CD = 0.9). Individual horizontal PFV profiles were determined using least squares linear regressions. F_0 represents the x-intercept of the FV relationship, while V_0 represents the y-intercept of the FV relationship.

2.7. Asymmetry analysis

The method proposed by Bishop et al. [23] was adopted for determining asymmetry. Using the data obtained from the isokinetic tests, the average extremity asymmetries were calculated utilizing the standard percentage difference equation. The equation is formulated as follows:

$$100 - (max\,value) * (min\,value)^{-1} + 100 \tag{10}$$

To ascertain the direction of asymmetry, an 'IF function' was incorporated into Microsoft Excel. This function, as suggested early research [23] identified which of the left and right extremities exhibited superior performance. The function formula is expressed as:

$$IF(left < right, 1, -1) \tag{11}$$

The inclusion of this function ensured that the magnitude of asymmetry remained unaltered despite variations in performance between different extremities.

2.8. Statistical analysis

SPSS version 22 on Windows 11 and GraphPad Prism version 10.2.1 analyzed the data. Normality was confirmed with the Shapiro-Wilks test. Variables with normal distribution are presented as mean (X) ± standard deviation (SD). Pearson Correlation Analysis assessed intra- and inter-limb strength asymmetry's relation to sprint kinematics and kinetics by Hopkins [24] is as follows: 0.0, small: 0.1–0.3, moderate: 0.3–0.5, high: 0.5–0.7, very high: 0.7–0.9, nearly perfect: 0.9, and perfect 1.0 relationships. Predictions were based on Intraclass Correlation Coefficients (ICCs) and their 95% confidence intervals, utilizing Koo and Li's classifications [25] for ICC values. Significance was set at $\alpha < 0.05$.

3. Results

Table 1 provides a summary of the anthropometric characteristics, body composition metrics, and maturity offset values separately presented for young team athletes: football, volleyball, and handball. The average values of variables such as age, body mass, body height, body fat percentage, years of training and maturity offset, along with their standard deviations, offer insights into the variability and distribution of these variables within each sports discipline.

Table 2 provides an overview of various parameters related to strength asymmetry, sprint kinematics and PFV profile parameters among the participants. Notably, the mean of intra and inter-limb asymmetries exhibit a wide range, varying from 8.62% to 11.90%, with minimum values of 0% and maximum values reaching up to 42.50%.

The association between intra and inter-limb strength asymmetry and sprint kinematics are presented in Graph 1. H60.s^{-1} demonstrated significant weak to moderate negative correlation with S_{30m} ($r = -0.27$, $p = 0.046$). There were also significant moderate positive correlations observed between H60.s^{-1} and LSV ($r = 0.43$, $p = 0.001$) as well as RSV ($r = 0.40$, $p = 0.002$). ILR60.s^{-1} exhibited significant moderate positive correlations with both LSV and RSV ($r = 0.38$, $p = 0.004$; $r = 0.37$, $p = 0.006$, respectively). A significant moderate positive correlation was identified between ILR180.s^{-1} and S_{5m} and S_{10m} ($r = 0.32$, $p = 0.017$; $r = 0.34$, $p = 0.012$, respectively). ILR180.s^{-1} demonstrated significant moderate negative correlations with LCT, RCT, LSL, RSL, LSV and RSV ($r = -0.37$, $p = 0.003$; $r = -0.39$, $p = 0.005$; $r = -0.45$, $p < 0.001$; $r = -0.46$, $p < 0.001$; $r = -0.34$, $p = 0.012$; $r = -0.30$, $p = 0.027$, respectively).

The association between intra- and inter-limb strength asymmetry and sprint kinetics are presented in Graph 2. H180.s^{-1} demonstrated significant moderate negative correlation with F_0 ($r = -0.45$, $p < 0.001$) and P_{max} ($r = -0.43$, $p < 0.001$). It also exhibited significant moderate positive correlation with FV_{slope} ($r = 0.44$, $p < 0.001$), RF_{max} ($r = 0.42$, $p < 0.001$) and D_{RF} ($r = 0.45$, $p < 0.001$). A significant moderate negative correlation was found between ILR180.s^{-1} and both F_0 ($r = -0.33$, $p = 0.009$) and P_{max} ($r = -0.33$, $p = 0.011$). ILR180.s^{-1} also demonstrated significant moderate positive correlations with FV_{slope} and D_{RF} ($r = 0.33$, $p = 0.011$; $r = 0.33$, $p = 0.011$, respectively).

4. Discussion

The study reveals weak to moderate correlations ($r = -46$ to 45) between asymmetry and sprint performance parameters among adolescent team athletes, including kinematic and kinetic aspects. Notably, while most participants had asymmetry levels below the 15% threshold considered functionally significant, some individuals exhibited asymmetry values up to 42%, indicating substantial variation. Optimal concentric hamstring-to-quadriceps isokinetic torque ratios typically range from 0.5 to 0.75. However, bilateral strength discrepancies exceeding 15% are rare and may suggest increased injury risk or incomplete rehabilitation [5,26]. This poses a concern for young team athletes, especially considering incomplete muscular and skeletal maturity, particularly for the muscle groups studied, given their importance in team sports and injury prevention.

The study observed that intra and inter isometric strength asymmetries were minimal at Q180°.s⁻1 (8.62%) and maximal at H180°.s⁻1 (11.90%). These small differences possibly explain the lack of significant high relationships between kinetic and kinematic variables. However, results align with prior research by [27], suggesting that when inter-limb strength disparities

Table 2
Descriptive statistics of intra- and inter-limb strength asymmetry, sprint kinematics, and power-force–velocity (PFV) profile parameters. Hamstring Strength at 60°.s^{-1} (H60°.s^{-1}) | Quadriceps Strength at 60°.s-1 (Q60°.s^{-1}) | Hamstring Strength at 180°.s^{-1} (H180°.s^{-1}) | Quadriceps Strength at 180°.s^{-1} (Q180°.s^{-1}).

Variables	X (95% CI)	SD	Min	Max
H60 .s^{-1}	10.18 (8.10 – 12.27)	8.92	0.0	40.80
Q60 .s^{-1}	9.05 (7.36 – 10.74)	7.25	0.0	42.50
H180 .s^{-1}	11.90 (9.83 – 13.97)	8.88	0.0	41.41
Q180 .s^{-1}	8.62 (7.17 – 10.07)	6.20	0.0	23.62
ILR60 .s^{-1}	9.99 (8.17 – 11.81	7.79	0.0	35.81
ILR180 .s^{-1}	10.84 (8.69 – 13.00)	9.24	0.0	41.29
F_0 (N/kg)	13.67 (12.74 – 14.60	3.91	7.15	32.68
V_0 (m/s)	7.73 (7.57 – 7.88	0.65	6.20	9.10
P_{max} (W/kg)	26.12 (24.58 – 27.66	6.49	14.19	56.59
Sfv (N/kg/m/s)	−1.80 (−1.95 – −1.66)	0.61	−4.72	−0.90
RF_{max} (%)	47.49 (46.56 – 48.41)	3.90	30	53
DRF (%)	−12.95 (−15.39 – −10.51)	10.30	−46.49	4.50
S_{5m} (s)	1.12 (1.10 – 1.14)	0.91	0.95	1.34
S_{10m} (s)	1.89 (1.87 – 1.92)	0.11	1.67	2.17
S_{20m} (s)	3.33 (3.27 – 3.40)	0.26	2.94	4.01
S_{30m} (s)	4.52 (4.46 – 4.58)	0.26	4.06	5.17
LCT (s)	0.15 (0.14 – 0.15)	0.01	0.12	0.20
RCT (s)	0.15 (0.14 – 0.15)	0.01	0.12	0.21
LFT (s)	0.09 (0.9 – 0.10)	0.01	0.07	0.15
RFT (s)	0.09 (0.09 – 0.10)	0.01	0.67	0.15
LSL (cm)	162.73 (158.75 – 166.70)	14.83	134.40	198.70
RSL (cm)	165.18 (161.13 – 169.24)	15.15	129.40	203
LSV (m.s^{-1})	6.91 (6.68 – 7.13)	0.84	1.47	7.87
RSV (m.s^{-1})	6.83 (6.56 – 7.10)	0.99	0.43	7.84

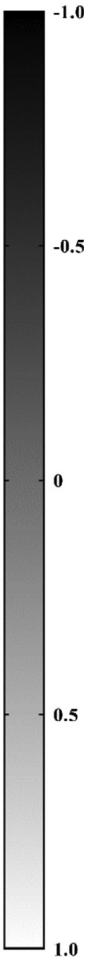
X: Mean; SD: Standard deviation; 95% CI: 95% Confidence interval. H60°.s^{-1}: Peak hamstring strength measured at an angular velocity of 60 degrees per second (absolute peak torque in Newton-meters). Q60°.s-1: Peak quadriceps strength measured at an angular velocity of 60 degrees per second (absolute peak torque in Newton-meters). H180°.s^{-1}: Peak hamstring strength measured at an angular velocity of 180 degrees per second (absolute peak torque in Newton-meters). Q180°.s^{-1}: Peak quadriceps strength measured at an angular velocity of 180 degrees per second (absolute peak torque in Newton-meters). ILR60°.s^{-1}: Intra-limb ratio of peak strength at 60 degrees per second, representing the difference in strength between dominant and non-dominant limbs. ILR180°.s^{-1}: Intra-limb ratio of peak strength at 180 degrees per second, representing the difference in strength between dominant and non-dominant limbs. S5m: Time to complete a 5-meter sprint (in seconds). S10m: Time to complete a 10-meter sprint (in seconds). S20m: Time to complete a 20-meter sprint (in seconds). S30m: Time to complete a 30-meter sprint (in seconds). LCT: Left leg contact time during sprinting (in seconds). RCT: Right leg contact time during sprinting (in seconds). LFT: Left leg flight time during sprinting (in seconds). RFT: Right leg flight time during sprinting (in seconds). LSL: Left step length during sprinting (in centimeters). RSL: Right step length during sprinting (in centimeters). LSV: Left step velocity during sprinting (in meters per second). RSV: Right step velocity during sprinting (in meters per second). F_0: Theoretical maximum force normalized to body mass (in Newtons per kilogram). V_0: Theoretical maximum velocity (in meters per second). Pmax: Theoretical maximum power normalized to body mass (in Watts per kilogram). FVslope: The slope of the force–velocity relationship, representing the rate of decrease in force as velocity increases. RFmax: The maximal value of the ratio of horizontal force to total force during sprinting (expressed as a percentage). DRF: The decrease in the ratio of horizontal force during sprinting, representing the decline in effective force production as speed increases (expressed as a percentage).

are below 15%, they may not impact linear sprinting performance significantly. Participation in sports emphasizing specific body sides can induce asymmetrical tissue changes. While some asymmetry is normal, prolonged unilateral loading during sports activities can lead to functional asymmetries [28]. The study's narrow age range implies that prolonged asymmetrical loading over years could potentially affect sprint kinematics and power FV profile. Hence, while some asymmetry is normal, prolonged unilateral loading may induce various functional or detrimental asymmetries.

While some strength asymmetry measures showed moderate positive correlations with step velocities (LSV and RSV), overall associations were not strong. Moderate correlations between strength asymmetry and short sprint distances (S5m and S10m) suggest limited impact on sprint acceleration. Enhanced acceleration skills in team sports confer strategic advantages, particularly in rapid bursts of short sprints [8]. Interestingly, increased asymmetry in inter-limb and intra and inter-limb ratios correlates with ILR60 .s-1 and ILR180 .s^{-1} (intra and inter-limb ratios). [17] found a positive correlation between hamstring-to-quadriceps muscle ratio and change of direction speed and linear sprint performance in elite soccer players under 21, suggesting that muscle disparities may enhance agility and speed performance.

The observed moderate positive correlations with certain kinetic parameters (e.g., FV slope and dynamic force ratio) suggest potential compensatory mechanisms at play. Statistically, despite moderate correlations between H180.s^{-1} and ILR180.s^{-1} isokinetic strength asymmetry, the fact that all components of the PFV profile exhibit significant relationships in these measurements can be considered a noteworthy finding. Furthermore, the significant association between both intra- and inter-limb asymmetries and PFV components at 180.s^{-1} is remarkable. Based on these findings, it could be posited that asymmetry in muscle function might manifest under conditions of low-speed force production, yet its direct correlation with sprint performance remains uncertain. However, the observed association between asymmetry and sprint power-FV components in assessments conducted at higher velocities, particularly at 180.s^{-1}, suggests that mechanical disparities stemming from increased muscular contraction speed could exhibit a more pronounced relationship with factors impacting sprint per-

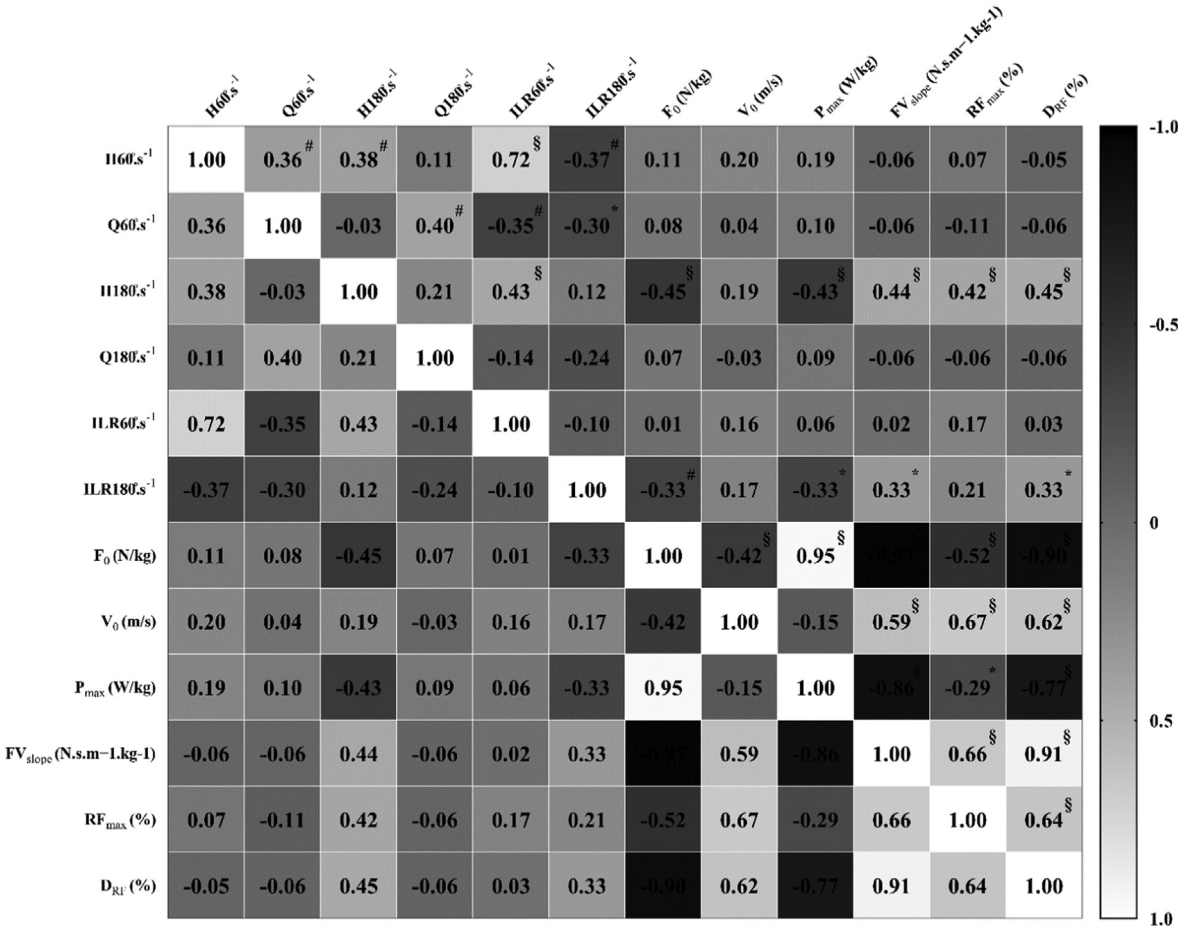
	H60.s^{-1}	Q60.s^{-1}	H180.s^{-1}	Q180.s^{-1}	ILR60.s^{-1}	ILR180.s^{-1}	S_{5m} (s)	S_{10m} (s)	S_{20m} (s)	S_{30m} (s)	LCT (s)	RCT (s)	LFT (s)	RFT (s)	LSL (cm)	RSL (cm)	LSV (m.s^{-1})	RSV (m.s^{-1})
H60.s^{-1}	1.00	0.36$^{\#}$	0.38$^{\#}$	0.11	0.72§	-0.37$^{\#}$	-0.13	-0.24	-0.18	-0.27*	-0.05	-0.04	0.04	0.04	0.10	0.23	0.43$^{\#}$	0.40$^{\#}$
Q60.s^{-1}	0.36	1.00	-0.03	0.40$^{\#}$	-0.35$^{\#}$	-0.30*	-0.10	-0.14	-0.12	-0.09	0.04	0.02	0.11	0.12	0.10	0.12	0.04	0.06
H180.s^{-1}	0.38	-0.03	1.00	0.21	0.43$^{\#}$	0.12	0.19	0.10	-0.03	0.03	-0.15	-0.11	0.06	0.06	-0.12	-0.05	0.15	0.15
Q180.s^{-1}	0.11	0.40	0.21	1.00	-0.14	-0.24	-0.13	-0.15	-0.15	-0.09	0.08	0.19	0.03	0.03	0.09	0.18	-0.16	-0.23
ILR60.s^{-1}	0.72	-0.35	0.43	-0.14	1.00	-0.10	-0.05	-0.12	-0.09	-0.22	-0.08	-0.08	-0.04	-0.04	0.07	0.12	0.38$^{\#}$	0.37$^{\#}$
ILR180.s^{-1}	-0.37	-0.30	0.12	-0.24	-0.10	1.00	0.32*	0.34*	-0.17	0.13	-0.37$^{\#}$	-0.39$^{\#}$	0.03	0.02	-0.45§	-0.46§	-0.34*	-0.30*
S_{5m} (s)	-0.13	-0.10	0.19	-0.13	-0.05	0.32	1.00	0.93§	0.19	0.59§	-0.12	-0.21	0.09	0.08	-0.50§	-0.49§	-0.12	-0.01
S_{10m} (s)	-0.24	-0.14	0.10	-0.15	-0.12	0.34	0.93	1.00	0.34*	0.80§	-0.02	-0.07	0.06	0.06	-0.54§	-0.53§	-0.28*	-0.19
S_{20m} (s)	-0.18	-0.12	-0.03	-0.15	-0.09	-0.17	0.19	0.34	1.00	0.53§	0.40$^{\#}$	0.45$^{\#}$	-0.09	-0.09	0.12	0.12	-0.18	-0.21
S_{30m} (s)	-0.27	-0.09	0.03	-0.09	-0.22	0.13	0.59	0.80	0.53	1.00	0.32*	0.35$^{\#}$	-0.07	-0.06	-0.32*	-0.31*	-0.37$^{\#}$	-0.34*
LCT (s)	-0.05	0.04	-0.15	0.08	-0.08	-0.37	-0.12	-0.02	0.40	0.32	1.00	0.93§	-0.10	-0.09	0.62§	0.58§	-0.06	-0.04
RCT (s)	-0.04	0.02	-0.11	0.19	-0.08	-0.39	-0.21	-0.07	0.45	0.35	0.93	1.00	-0.19	-0.17	0.56§	0.59§	-0.05	-0.12
LFT (s)	0.04	0.11	0.06	0.03	-0.04	0.03	0.09	0.06	-0.09	-0.07	-0.10	-0.19	1.00	1.00§	-0.09	-0.09	0.06	0.14
RFT (s)	0.04	0.12	0.06	0.03	-0.04	0.02	0.08	0.06	-0.09	-0.06	-0.09	-0.17	1.00	1.00	-0.09	-0.08	0.05	0.13
LSL (cm)	0.10	0.10	-0.12	0.09	0.07	-0.45	-0.50	-0.54	0.12	-0.32	0.62	0.56	-0.09	-0.09	1.00	0.84§	0.15	0.13
RSL (cm)	0.23	0.12	-0.05	0.18	0.12	-0.46	-0.49	-0.53	0.12	-0.31	0.58	0.59	-0.09	-0.08	0.84	1.00	0.25	0.18
LSV (m.s^{-1})	0.43	0.04	0.15	-0.16	0.38	-0.34	-0.12	-0.28	-0.18	-0.37	-0.06	-0.05	0.06	0.05	0.15	0.25	1.00	0.95§
RSV (m.s^{-1})	0.40	0.06	0.15	-0.23	0.37	-0.30	-0.01	-0.19	-0.21	-0.34	-0.04	-0.12	0.14	0.13	0.13	0.18	0.95	1.00



Graph 1. Association between intra and inter-limb strength asymmetry and sprint kinematics. H60 .s^{-1}: Hamstrings 60 .s^{-1}; Q60 .s^{-1}: Quadriceps 60 .s^{-1}; H180 .s^{-1}: Hamstrings 180 .s^{-1}; Q180 .s^{-1}: Quadriceps 180 .s^{-1}; ILR60 .s^{-1}: Intra-limb Ratio 60 .s^{-1}; ILR180 .s^{-1}: Intra-limb Ratio 180 .s^{-1}; S_{5m}: 5 m sprint time; S_{10m}: 10 m sprint time; S_{20m}: 20 m sprint time; S_{30m}: 30 m sprint time; LCT: Left contact time; RCT: Right contact time; LFT: Left flight time; RFT: Right flight time; LSL: Left step length; RSL: Right step length; LSV: Left step velocity; RSV: Right step velocity. *= $p < 0.05$; # = $p < 0.01$; § = $p < 0.001$.

formance. Moreover, the adverse impact of limb asymmetries on the FV profile of youth athletes is notable in relation to the attenuation of theoretical maximum force (F_0). Sprinting entails athletes overcoming initial resistance to attain high velocity from a stationary position, necessitating substantial horizontal force and power exertion as running speed escalates [13]. Studies have consistently demonstrated a positive correlation between maximal force capabilities and running performance parameters such as sprint speed, stride length, and running economy [2]. Moreover, adequate maximal force capacity enables runners to withstand the repetitive impact forces associated with running, reducing the risk of fatigue-related injuries and enhancing overall durability [8].

The study has limitations: a small sample of youth team athletes, uneven sample distribution, and exclusive focus on male athletes. Generalizability to broader athlete populations is thus limited, requiring validation across diverse samples. Future research should include female athletes for broader inclusivity. The study used cross-sectional design and emphasized the importance of repeated isokinetic measurements for evaluating extremity strength asymmetry and training effectiveness [1,29]. Regular monitoring of extremity strength asymmetry is advised for performance evaluation [5]. These findings are relevant for sports professionals like physicians, coaches, trainers, and physiotherapists. Addressing extremity strength imbalances in training plans is crucial for injury prevention and performance optimization [30]. advancing athlete management and injury risk reduction.



Graph 2. Association between intra and inter-limb strength asymmetry and sprint power-force–velocity profile parameters. H60 $.s^{-1}$: Hamstrings 60 $.s^{-1}$; Q60 $.s^{-1}$: Quadriceps 60 $.s^{-1}$; H180 $.s^{-1}$: Hamstrings 180 $.s^{-1}$; Q180 $.s^{-1}$: Quadriceps 180 $.s^{-1}$; ILR60 $.s^{-1}$: Intra-limb Ratio 60 $.s^{-1}$; ILR180 $.s^{-1}$: Intra-limb Ratio 180 $.s^{-1}$; F_0: Theoretical maximum force; V_0: Theoretical maximum speed; P_{max}: Theoretical maximum power; FV_{slope}: Slope of force–velocity relationship; RF_{max}: Maximal value of ratio of horizontal force; D_{RF}: Decrease in the ratio of horizontal force. *= $p < 0.05$; # = $p < 0.01$; § = $p < 0.001$.

5. Conclusion

These findings challenge conventional beliefs regarding the impact of strength asymmetry on sprint biomechanics, suggesting that the relationship is nuanced and multifaceted. For instance, the correlation coefficients ranged from −0.46 to 0.45 for various strength asymmetry measures (such as H60.s-1, ILR60.s-1, ILR180.s-1) and sprint kinetics parameters (including S_{5m}, S_{10m}, S_{30m}, LSV, RSV, LCT, RCT, LSL, RSL, F_0, P_{max}, FV_{slope}, RF_{max}, and D_{RF}). While some correlations were observed, they were moderate and inconsistent across different parameters, highlighting the need for further research to elucidate underlying mechanisms.

CRediT authorship contribution statement

İzzet Ince: Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Erkan Tortu:** Writing – review & editing, Visualization, Data curation.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

[1] Didehdar D, Kharazinejad A. The association between sprint speed test and isokinetic knee strength in healthy male volleyball players. Inter J Sport Stud Health 2021;4(2).
[2] Ciacci S, Di Michele R, Merni F. Kinematic analysis of the braking and propulsion phases during the support time in sprint running. Gait Posture 2010;31(2):209–12.
[3] Fox KT, Pearson LT, Hicks KM. The effect of lower inter-limb asymmetries on athletic performance: A systematic review and meta-analysis. PLoS One 2023;18(6):e0286942.
[4] Kabacinski J, Szozda PM, Mackala K, Murawa M, Rzepnicka A, Szewczyk P, et al. Relationship between isokinetic knee strength and speed, agility, and explosive power in elite soccer players. Int J Environ Res Public Health 2022;19(2):671.
[5] Bishop C, Turner A, Read P. Effects of inter-limb asymmetries on physical and sports performance: A systematic review. J Sports Sci 2018;36(10):1135–44.
[6] Exell T, Irwin G, Gittoes M, Kerwin D. Strength and performance asymmetry during maximal velocity sprint running. Scand J Med Sci Sports 2017;27(11):1273–82.
[7] Exell T. Lower-limb biomechanical asymmetry in maximal velocity sprint running. Cardiff Metropolitan University; 2010.
[8] Macadam P, Simperingham KD, Cronin JB. Acute kinematic and kinetic adaptations to wearable resistance during sprint acceleration. J Strength Cond Res 2017;31(5):1297–304.
[9] Samozino P, Peyrot N, Edouard P, Nagahara R, Jimenez-Reyes P, Vanwanseele B, et al. Optimal mechanical force-velocity profile for sprint acceleration performance. Scand J Med Sci Sports 2022;32(3):559–75.
[10] Alt T, Komnik I, Severin J, Nodler YT, Benker R, Knicker AJ, et al. Swing Phase Mechanics of Maximal Velocity Sprints—Does Isokinetic Lower-Limb Muscle Strength Matter? Int J Sports Physiol Perform 2021;16(7):974–84.
[11] Nagahara R, Murata M. Inertial Measurement Unit Based Hip Flexion Strength-Power Test for Sprinters. Front Sports Active Living 2020;2:571523.
[12] Shealy MJ, Callister R, Dudley GA, Fleck SJ. Human torque velocity adaptations to sprint, endurance, or combined modes of training. Am J Sports Med 1992;20(5):581–6.
[13] Samozino P, Rabita G, Dorel S, Slawinski J, Peyrot N, Saez de Villarreal E, et al. A simple method for measuring power, force, velocity properties, and mechanical effectiveness in sprint running. Scand J Med Sci Sports 2016;26(6):648–58.
[14] Keogh JA, Waddington EE, Masood Z, Mahmood S, Palanisamy AC, Ruder MC, et al. Monitoring lower limb biomechanical asymmetry and psychological measures in athletic populations—A scoping review. Scand J Med Sci Sports 2023;33(11):2125–48.
[15] Exell T, Gittoes M, Irwin G, Kerwin D. Asymmetry in sprint running: Strength and performance interactions. ISBS-Conference Proceedings Archive. 2015.
[16] Prvulović N, Čoh M, Čular D, Tomljanović M, Sporiš G, Fišer SŽ. Countermovement jump in female sprinters: kinetic parameters and asymmetry. Symmetry 2022;14(6):1130.
[17] Coratella G, Beato M, Schena F. Correlation between quadriceps and hamstrings inter-limb strength asymmetry with change of direction and sprint in U21 elite soccer-players. Hum Mov Sci 2018;59:81–7.
[18] Simonsmeier BA, Androniе M, Buecker S, Frank C. The effects of imagery interventions in sports: A meta-analysis. Int Rev Sport Exerc Psychol 2021;14(1):186–207.
[19] Kozieł SM, Malina RM. Modified maturity offset prediction equations: validation in independent longitudinal samples of boys and girls. Sports Med 2018;48:221–36.
[20] Ulupınar S, İnce İ. Prediction of competition performance via commonly used strength-power tests in junior female weightlifters. Isokinet Exerc Sci 2021;29(3):309–17.
[21] İnce İ, Ulupınar S, Özbay S. Body composition isokinetic knee extensor strength and balance as predictors of competition performance in junior weightlifters. Isokinet Exerc Sci 2020;28(2):215–22.
[22] Pardos-Mainer E, Casajús JA, Gonzalo-Skok O. Reliability and sensitivity of jumping, linear sprinting and change of direction ability tests in adolescent female football players. Sci Med Football 2019;3(3):183–90. doi: https://doi.org/10.1080/24733938.2018.1554257.
[23] Bishop C, Coratella G, Beato M. Intra-and inter-limb strength asymmetry in soccer: A comparison of professional and under-18 players. Sports 2021;9(9):129.
[24] Hopkins W. A scale of magnitudes for effect statistics. A New View. Stat 2017;2002.
[25] Koo TK, Li MY. A guideline of selecting and reporting intraclass correlation coefficients for reliability research. J Chiropr Med 2016;15(2):155–63.
[26] Parpa K, Michaelides M. Anterior-posterior and inter-limb lower body strength asymmetry in soccer, basketball, futsal, and volleyball players. Medicina 2022;58(8):1080.
[27] Lockie RG, Schultz AB, Jeffriess MD, Callaghan SJ. The relationship between bilateral differences of knee flexor and extensor isokinetic strength and multi-directional speed. Isokinet Exerc Sci 2012;20(3):211–9.
[28] Hart NH, Nimphius S, Weber J, Spiteri T, Rantalainen T, Dobbin M. et al., Musculoskeletal asymmetry in football athletes: a product of limb function over time (2016).
[29] Bishop C, Read P, Lake J, Chavda S, Turner A. Interlimb asymmetries: Understanding how to calculate differences from bilateral and unilateral tests. Strength Condit J 2018;40(4):1–6.
[30] Oleksy Ł, Mika A, Sulowska-Daszyk I, Kielnar R, Dzięcioł-Anikiej Z, Zyznawska J, et al. The evaluation of asymmetry in isokinetic and electromyographic activity (sEMG) of the knee flexor and extensor muscles in football players after ACL rupture reconstruction and in the athletes following mild lower-limb injuries. J Clin Med 2023;12(3):1144.